

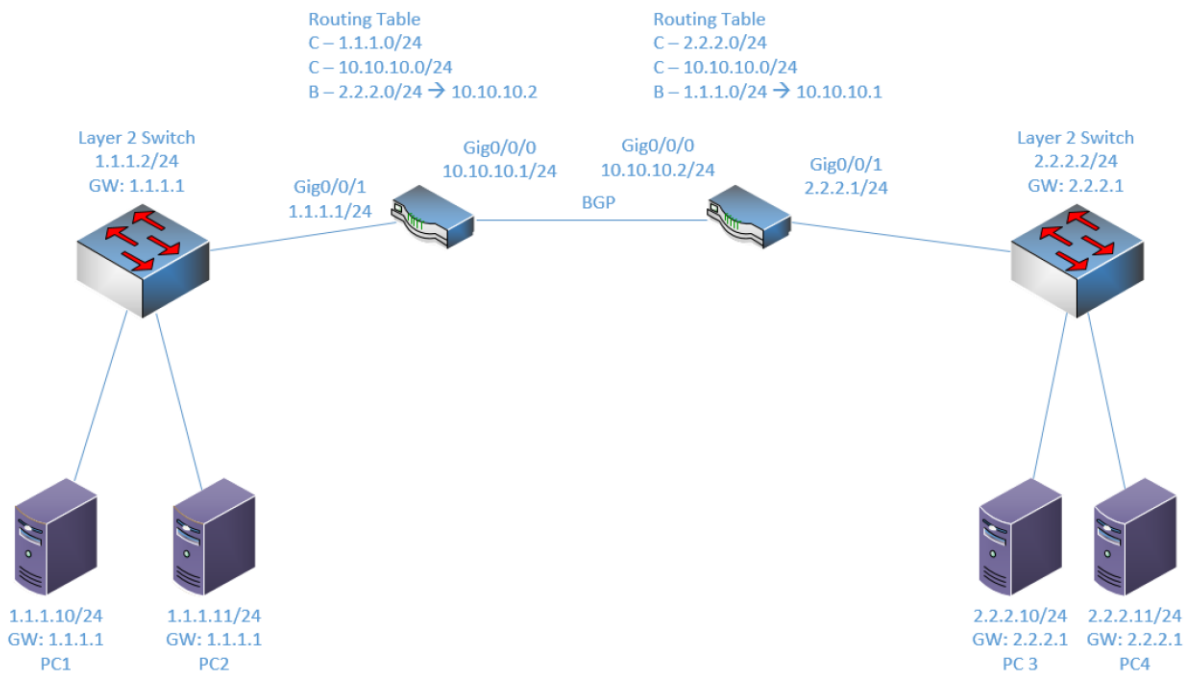
# Lab 11

## Cisco IOS device hardening

### Objective:

The purpose of this lab is for the student to understand configuration and implement remote access VPN using Cisco ASA and AnyConnect Secure Mobility Client

### Diagram:



### Components:

- 4 – PCs
- 1 – Cisco 3650 Layer 2 Switches
- 2 – Cisco 4321 ISR Routers

Useful commands for this lab

This lab will use Cisco IOS devices to configure secure Management, Control and Data plane options.

**Procedure:**

1. Connect all PCs, switches and routers as shown in the diagram above.
2. Configure the IP addresses on the router interfaces as shown above. The routing tables and interface IP addresses are all shown above.
3. Configure IP addresses, subnet masks and gateways on PCs.
4. Confirm PCs on same network can ping each other and their gateway.
5. Configure BGP between Route 1 and Router 2

a. Router 1:

```
router bgp 1
  neighbor 10.10.10.2 remote-as 2
  neighbor 10.10.10.2 password Netcom230@#)
  neighbor 10.10.10.2 ttl-security hops 100
  network 1.1.1.0 mask 255.255.255.0
```

b. Router 2:

```
router bgp 2
  neighbor 10.10.10.1 remote-as 1
  neighbor 10.10.10.1 password Netcom230@#)
  neighbor 10.10.10.1 ttl-security hops 100
  network 2.2.2.0 mask 255.255.255.0
```

6. Confirm BGP is established between the two devices. Use the “show ip bgp neighbor” and “show ip bgp summary” commands. Paste the output below.

Router 1:

```

Router#show ip bgp neighbor
BGP neighbor is 10.10.10.2, remote AS 2, external link
  BGP version 4, remote router ID 10.10.10.2
  BGP state = Established, up for 00:05:07
  Last read 00:00:35, last write 00:00:30, hold time is 180, keepalive interval
  is 60 seconds
  Neighbor sessions:
    1 active, is not multisession capable (disabled)
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Four-octets ASN Capability: advertised and received
    Address family IPv4 Unicast: advertised and received
    Enhanced Refresh Capability: advertised and received
    Multisession Capability:
    Stateful switchover support enabled: NO for session 1
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

      Sent      Rcvd
Opens:          1          1
Notifications:  0          0
Updates:        2          2
Keepalives:     7          7
Route Refresh:  0          0
Total:         10         10
Do log neighbor state changes (via global configuration)
Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast
Session: 10.10.10.2
BGP table version 3, neighbor version 3/0
Output queue size : 0
Index 1, Advertise bit 0
1 update-group member
Slow-peer detection is disabled
Slow-peer split-update-group dynamic is disabled

      Sent      Rcvd
Prefix activity: ----
Prefixes Current:    1          1 (Consumes 136 bytes)
Prefixes Total:      1          1
Implicit Withdraw:    0          0
Explicit Withdraw:    0          0
Used as bestpath:     n/a        1
Used as multipath:     n/a        0
Used as secondary:     n/a        0

      Outbound   Inbound
Local Policy Denied Prefixes: -----
  Bestpath from this peer:          1          n/a
  Total:                          1          0
Number of NLRI's in the update sent: max 1, min 0
Last detected as dynamic slow peer: never
Dynamic slow peer recovered: never
Refresh Epoch: 1
Last Sent Refresh Start-of-rib: never
Last Sent Refresh End-of-rib: never
Last Received Refresh Start-of-rib: never
Last Received Refresh End-of-rib: never

```

```

Refresh activity:
Refresh Start-of-RIB      0      0
Refresh End-of-RIB        0      0

Address tracking is enabled, the RIB does have a route to 10.10.10.2
Route to peer address reachability Up: 1; Down: 0
  Last notification 00:07:32
Connections established 1; dropped 0
Last reset never
External BGP neighbor may be up to 100 hops away.
External BGP neighbor NOT configured for connected checks (multi-hop no-disabl
e-connected-check)
Interface associated: (none) (peering address in same link)
Transport(tcp) path-mtu-discovery is enabled
Graceful-Restart is disabled
SSO is disabled
Connection state is ESTAB, I/O status: 1, unread input bytes: 0
Connection is ECN Disabled, Mininum incoming TTL 155, Outgoing TTL 255
Local host: 10.10.10.1, Local port: 38607
Foreign host: 10.10.10.2, Foreign port: 179
Connection tableid (VRF): 0
Maximum output segment queue size: 50

Enqueued packets for retransmit: 0, input: 0  mis-ordered: 0 (0 bytes)

Event Timers (current time is 0x121C1F):
Timer           Starts    Wakeups      Next
Retrans         9         0            0x0
TimeWait        0         0            0x0
AckHold         8         5            0x0
SendWnd         0         0            0x0
KeepAlive       0         0            0x0
GiveUp          0         0            0x0
PmtuAger        1         0            0x169380
DeadWait        0         0            0x0
Linger          0         0            0x0
ProcessQ        0         0            0x0

ss: 3481314927  snduna: 3481315195  sndnxt: 3481315195
rs: 571113898  rcvnxt: 571114166

sndwnd: 16117  scale:      0  maxrcvwnd: 16384
rcvwnd: 16117  scale:      0  delrcvwnd: 267

SRTT: 699 ms, RTTO: 2656 ms, RTV: 1957 ms, KRTT: 0 ms
minRTT: 1 ms, maxRTT: 1000 ms, ACK hold: 200 ms
uptime: 307297 ms, Sent idletime: 30792 ms, Receive idletime: 30590 ms
Status Flags: active open
Option Flags: nagle, path mtu capable, md5
IP Precedence value : 6

Datagrams (max data segment is 1460 bytes):
Rcvd: 17 (out of order: 0), with data: 9, total data bytes: 267
Sent: 18 (retransmit: 0, fastretransmit: 0, partialack: 0, Second Congestion: 0)
  with data: 9, total data bytes: 267

Packets received in fast path: 0, fast processed: 0, slow path: 0
fast lock acquisition failures: 0, slow path: 0
TCP Semaphore      0x7F4EA29FECF0  FREE

Router#

```

```
Router#show ip bgp summary
BGP router identifier 10.10.10.1, local AS number 1
BGP table version is 3, main routing table version 3
2 network entries using 496 bytes of memory
2 path entries using 272 bytes of memory
2/2 BGP path/bestpath attribute entries using 560 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1352 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State
/PfxRcd
10.10.10.2    4        2     12     12       3    0   0 00:07:06
  1
Router#
```

Router 2:

```

Router#show ip bgp neighbor
BGP neighbor is 10.10.10.1, remote AS 1, external link
  BGP version 4, remote router ID 10.10.10.1
  BGP state = Established, up for 00:01:12
  Last read 00:00:17, last write 00:00:15, hold time is 180, keepalive interval
  is 60 seconds
  Neighbor sessions:
    1 active, is not multisession capable (disabled)
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Four-octets ASN Capability: advertised and received
    Address family IPv4 Unicast: advertised and received
    Enhanced Refresh Capability: advertised and received
    Multisession Capability:
    Stateful switchover support enabled: NO for session 1
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

              Sent      Rcvd
Opens:          1        1
Notifications:  0        0
Updates:        2        2
Keepalives:     3        3
Route Refresh:  0        0
Total:          6        6

Do log neighbor state changes (via global configuration)
Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast
  Session: 10.10.10.1
  BGP table version 5, neighbor version 5/0
  Output queue size : 0
  Index 1, Advertise bit 0
  1 update-group member
  Slow-peer detection is disabled
  Slow-peer split-update-group dynamic is disabled

              Sent      Rcvd
Prefix activity:
  ----
Prefixes Current:    1        1 (Consumes 136 bytes)
Prefixes Total:      1        1
Implicit Withdraw:    0        0
Explicit Withdraw:    0        0
Used as bestpath:     n/a      1
Used as multipath:     n/a      0
Used as secondary:     n/a      0

              Outbound    Inbound
Local Policy Denied Prefixes:  -----
Bestpath from this peer:      1        n/a
Total:                        1        0
Number of NLRI in the update sent: max 1, min 0
Last detected as dynamic slow peer: never
Dynamic slow peer recovered: never
Refresh Epoch: 1
Last Sent Refresh Start-of-rib: never

```

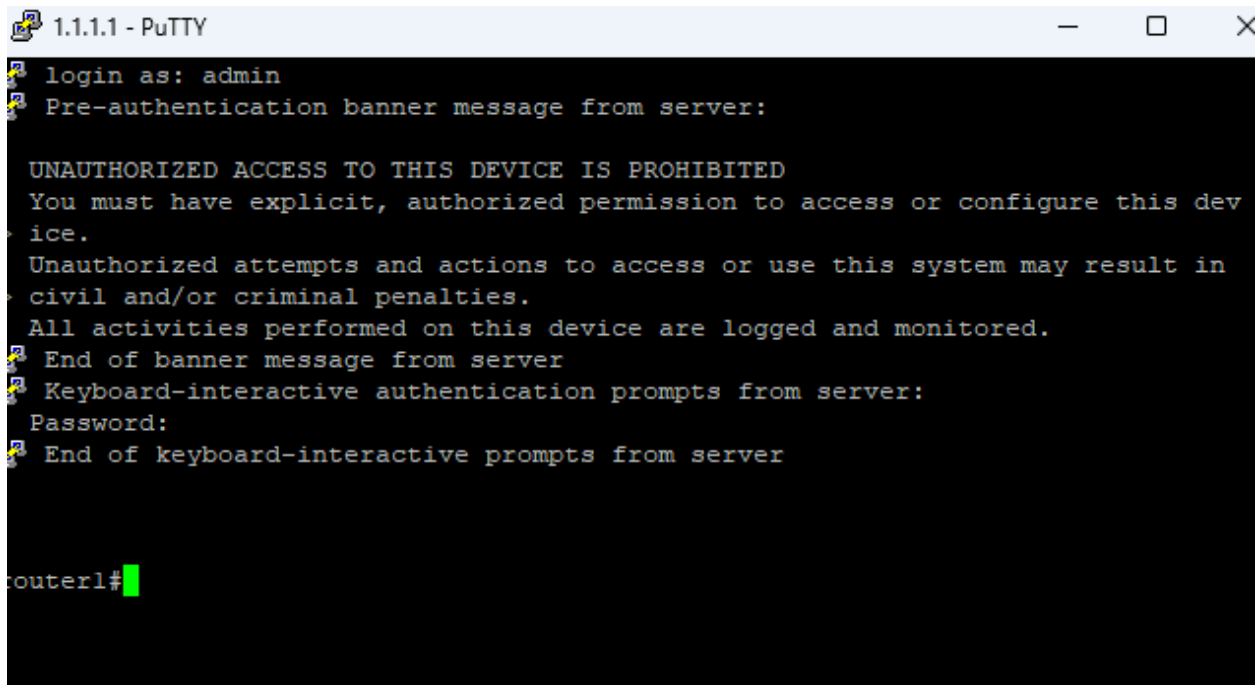
7. Once you're able to ping between the devices, configure SSH on both routers securely using the lab commands document included.
  - a. Configure the SSH ACL so that 1.1.1.10 and 2.2.2.10 can SSH into both routers, but 1.1.1.11 and 2.2.2.11 are both blocked. Remember, there is an implicit deny at the end of your ACLs so

you only need two entries on each router to permit the two endpoints ending with .10.

8. Configure a login banner using the lab commands doc with the following message:

```
banner login #
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED
You must have explicit, authorized permission to access or
configure this device.
Unauthorized attempts and actions to access or use this system
may result in civil and/or
criminal penalties.
All activities performed on this device are logged and monitored.
#
```

9. Use Putty to log into each router from the workstations ending with .10.
  - a. Screenshot your banner on one of the routers and paste below

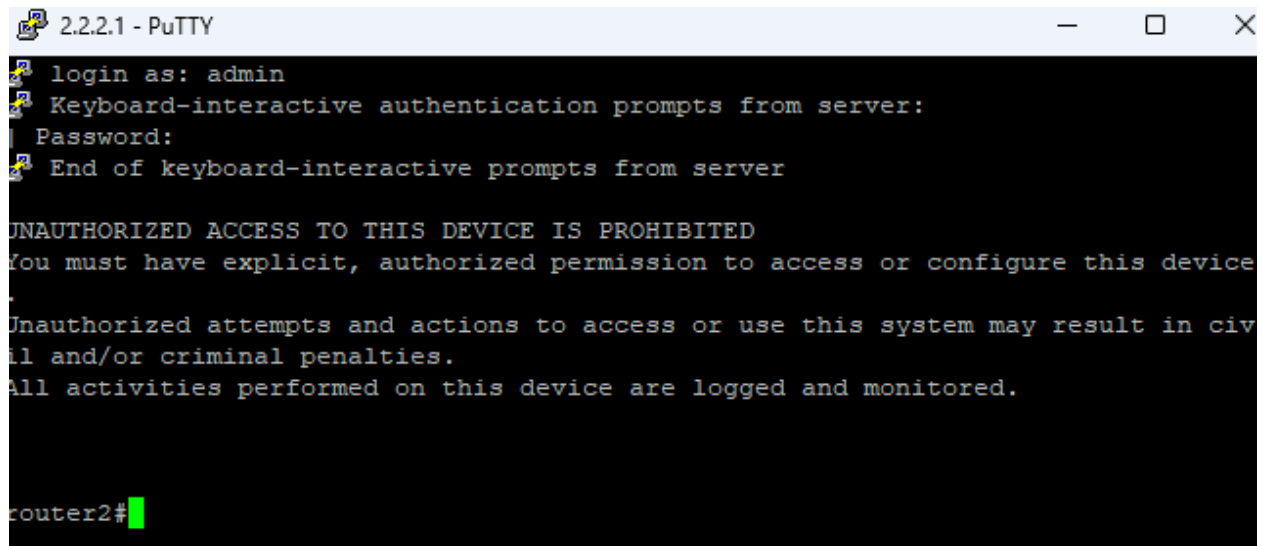


The screenshot shows a PuTTY terminal window titled "1.1.1.1 - PuTTY". The terminal output displays the login banner configuration and the login process. The banner message is: "UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED. You must have explicit, authorized permission to access or configure this device. Unauthorized attempts and actions to access or use this system may result in civil and/or criminal penalties. All activities performed on this device are logged and monitored." The user logs in as "admin" and the prompt changes to "router1#".

```
1.1.1.1 - PuTTY
login as: admin
Pre-authentication banner message from server:

UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED
You must have explicit, authorized permission to access or configure this dev
ice.
Unauthorized attempts and actions to access or use this system may result in
civil and/or criminal penalties.
All activities performed on this device are logged and monitored.
End of banner message from server
Keyboard-interactive authentication prompts from server:
Password:
End of keyboard-interactive prompts from server

router1#
```



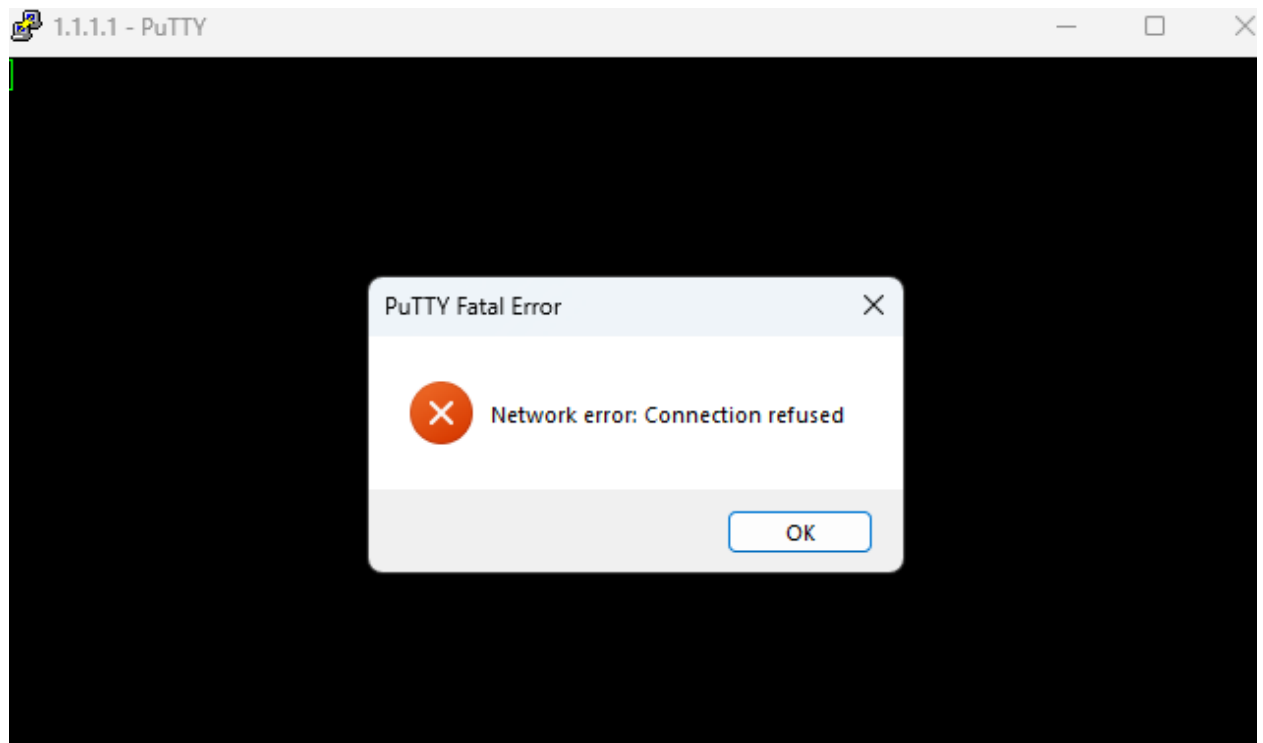
```
2.2.2.1 - PuTTY
login as: admin
Keyboard-interactive authentication prompts from server:
Password:
End of keyboard-interactive prompts from server

UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED
You must have explicit, authorized permission to access or configure this device
.
Unauthorized attempts and actions to access or use this system may result in civil and/or criminal penalties.
All activities performed on this device are logged and monitored.

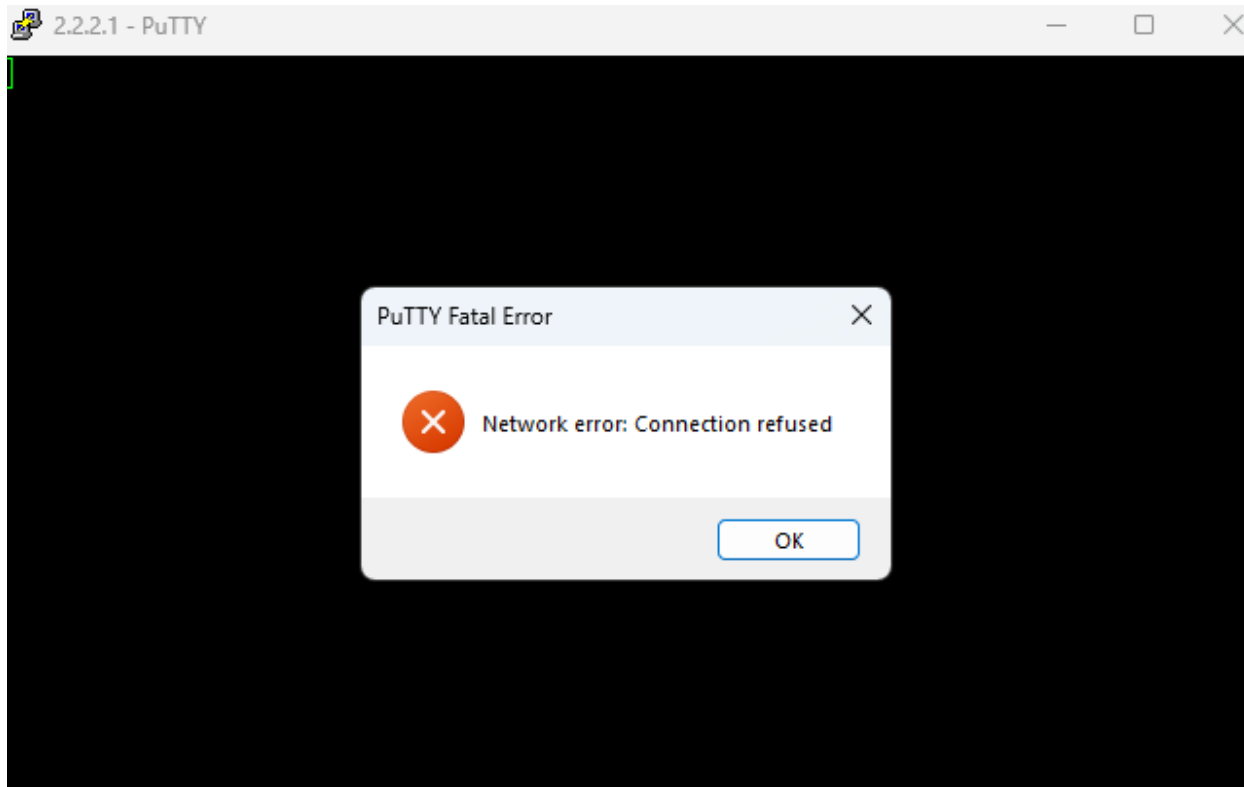
router2#
```

10. Try to log into the devices from the workstations ending with .11. Were you successful?

a. This should be denied. Paste the rejection message below







11. Next, add the control plane security options to the interfaces.

```
interface gig 0/0/0
  no ip redirects
  no ip unreachable
  no ip proxy-arp
```

12. These are a little tougher to prove they are working, so just take a screen shot of your interface config and paste below.

```
router1(config)#do show run int g0/0/0
Building configuration...

Current configuration : 147 bytes
!
interface GigabitEthernet0/0/0
 ip address 10.10.10.1 255.255.255.0
 no ip redirects
 no ip unreachable
 no ip proxy-arp
 negotiation auto
end

router1(config)#
```

```
router2(config)#do show run int g0/0/0
Building configuration...

Current configuration : 147 bytes
!
interface GigabitEthernet0/0/0
 ip address 10.10.10.2 255.255.255.0
 no ip redirects
 no ip unreachable
 no ip proxy-arp
 negotiation auto
end

router2(config)#
```

13. Next, configure the data plane security options listed in the command sheet. Disable ip options, ip source-route and ip redirects. Create an ACL to block fragmented packets. and enable uRPF. Take a screen shot of these config options and paste below.

```
interface GigabitEthernet0/0/0
 ip address 10.10.10.1 255.255.255.0
 no ip redirects
 no ip unreachable
 no ip proxy-arp
 ip verify unicast source reachable-via any
 ip access-group ACL-TRANSIT-IN in
 negotiation auto
!
interface GigabitEthernet0/0/1
 ip address 1.1.1.1 255.255.255.0
 negotiation auto
!
router bgp 1
 bgp log-neighbor-changes
 network 1.1.1.0 mask 255.255.255.0
 neighbor 10.10.10.2 remote-as 2
 neighbor 10.10.10.2 password Netcom230@#)
 neighbor 10.10.10.2 ttl-security hops 100
!
ip forward-protocol nd
no ip http server
ip http secure-server
ip route 172.20.1.0 255.255.255.0 1.1.1.2
!
```

```
ip access-list standard Permit_SSH
 permit 2.2.2.10
 permit 1.1.1.10
!
ip access-list extended ACL-TRANSIT-IN
 deny   tcp any any fragments
 deny   udp any any fragments
 deny   icmp any any fragments
 deny   ip any any fragments
 permit ip any any
!

control-plane
!
banner login ^C
UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED
You must have explicit, authorized permission to access or configure this device
.
Unauthorized attempts and actions to access or use this system may result in civil and/or criminal penalties.
All activities performed on this device are logged and monitored.
^C
!
line con 0
 transport input none
 stopbits 1
line vty 0 4
 access-class Permit_SSH in
 login local
 transport input ssh
!
```

- 14.If you would like to have a discussion about any of these options and their use cases, feel free to see me during class. I can find additional material and lab examples if you'd like.
- 15.Draw the lab diagram below (PCs, firewall and switch connections) Label the IP addresses, subnet masks, and gateways used by each device. Label the routing tables next to the routers.

